Chapter Eight

Fear of Machines

Machine Liberation

Many people fear a “takeover” by machines, but there has never been a single act or plan by machines to hurt anyone. (Unfortunately, the same cannot be said of human beings.) Humans, not machines, use nerve gas and missiles to destroy. Even automobile and aircraft accidents are mostly caused by human error rather than by mechanical failures.

Many people fear rapid technological developments, especially automated and cybernated machines, replacing human beings. To be fair, some of these fears are justified in a monetary system where rapid increases in production technology require fewer workers.

Some distrust a computerized society and fear possible failures of the machines. They worry that technology will make us more like machines, driving us toward uniformity, and resulting in the loss of individuality, freedom of choice, and privacy.

In defense of machines, there is no evidence of machines acting against human beings on their own, except in science fiction stories. Humans program machines and direct their use. It is not machines that are to be feared; it is the misuse and misdirection of these machines that threaten humankind. We must not forget that the bombing of cities, the use of nerve gas, prisons, death camps, and torture chambers have all been managed and operated by human beings, not machines. Even atomic weapons and guided missiles were built and directed by people. People pollute the environment -- our air, oceans, and rivers. The use and sale of harmful drugs, the distortion of truth, bigotry, and racial hatred are all part of flawed human systems and false indoctrination, and hardly characteristic of machines.

Machines are not the danger. We are. As long as we do not take responsibility for our relationship to fellow human beings and the intelligent management of our planetary resources, we are the greatest danger to the planet. If there ever were a conflict between people and machines, we know who would start it!

Science and technology create none of our problems. Our problems arise from human abuse and misuse of other people, the environment, and technology. In a more humane civilization, machines are used to shorten the workday, increase the availability of goods and services, and lengthen vacation time. New technology is used to raise the standard of living for everyone, and in this way, increasing the use of machine technology serves to benefit all people.

Chapter Nine

Cities in the Sea

The Ocean Frontiers

The web of life on our planet is supported by the hydro-cycle, that great variation of forms of water, which are part of the planetary circulation: the oceans, snow, ice, rain, lakes, groundwater, and aquifers. This constantly renewed circulation, powered by the heat of the sun, the rotation of the earth, and Coriolis forces, supports the entire life cycle, including humankind.

People often speak of underdeveloped land areas, but rarely of the greatest undeveloped natural resources on the planet, which are the world's oceans. Exploration and development of the oceans must be carried out with the utmost care. Although humans have used the oceans of the world for thousands of years as a source of food and transportation, we are just beginning to recognize the enormous potential and diversity of this relatively untapped resource. The oceans offer an almost limitless environment for food, energy production, transportation, minerals, pharmaceuticals, and much more.

In the past there was little regard for ocean life, which is essential to all life on Earth. We would survive and advance more easily as a species if we take seriously the reclamation of our oceans.

Past Abuses of the Ocean Environment

In August of 1970, the U. S. Army deliberately dumped containers holding 67 tons of nerve gas into the Atlantic Ocean. Worse, the dumping ground was close to a main artery in this life support system, the Gulf Stream, which makes the clean-up all the more urgent. The Navies of the world, the fishing fleets, cruise lines, and many coastal cities casually use the ocean as both trashcan and toilet.

The lack of adequate sanitation is one of the biggest threats to human health. It results in ill-health, disease, and death related to pollution of coastal waters. South Asia alone has 825 million people who live on the

coast without basic sanitation facilities. It isn't hard to understand why the levels of untreated sewage in South Asia's Coastal waters are the highest in the world. This, besides being a health risk to people, creates toxic algal blooms that cause mass kills of fish, wildlife and coral reefs. (4) Page 28 Transforming the Global Biosphere: Twelve Futuristic Strategies by Elliott Maynard, Ph.D.

Destructive environmental practices are numerous. Large commercial trawlers damage the sea-floor environment on a massive global scale. Their nets crush or bury seabed organisms, destroying their food and nursery grounds. This ecosystem is crucial for replenishing marine seafood stocks. (5) Ibid, page 70.

This process does more damage to the sea bed than clear cutting of the forests does to the Earth's surface. A single pass kills from 5 to 20 percent of seafloor animals, and this goes on twenty four hours-a-day seven days-a-week, year round on a global scale. (6) Ibid, Page 70-71.

Mismanagement of run-off has created huge lifeless areas in the Gulf of Mexico where the Mississippi River drains. Destructive business practices have over fished the waters to the point where most of the larger more productive reproducing fish are close to extinction. Throughout the world, major marine species and the coral reefs that nurture them are rapidly disappearing, but not naturally, or because their death in any way prolongs our way of life. On the contrary, these extinctions endanger us and derive from our own arrogance and ignorance. Even to the most complex living ecology, we act as predators.

New Respect of the Web of Life

With the resource based economy comes new value systems. Since no one gains financially through the wasteful practices of the past, the main objective is to reclaim and maintain a healthy and productive environment. If the oceans are intelligently managed, they can easily supply more than enough resources to feed the world's hungry. Billions could depend on the sea, where life is abundant and varied, for their primary source of protein. Although the overwhelming majority of sea life dwells near the surface, in the chill murky depths miles below where even sunlight never ventures, life abounds despite fantastic pressures and temperatures. In near-freezing temperatures, boiling vents of toxic gasses support a wide variety of sea life that remains to be studied.

Great rivers, called currents, cross the oceans of the planet, set in motion by Earth's rotation. These immense oceanic currents travel at varying

speeds, at different depths, and even in opposite directions. It is estimated that the Gulf Stream carries about 30 million cubic meters of water per second past Miami, Florida. This is more than five times the combined flow of all the fresh water rivers of the world.

By harnessing this potential energy, it is estimated that close to a thousand million watts on a 24-hour basis -- or as much as two large nuclear plants – could be generated, without environmental contamination or radiation danger.

In addition, powerful winds, waves, and currents provide us with enormous potential sources of electric power. Energy "crops" can be farmed from biomass by converting waste organic materials into gaseous or liquid fuels. Additional energy can be obtained from fermentation. Imagine a pile of decaying food and other organic matter. This pile of biomass gives off heat and gasses. This potential source of energy can be harnessed and used with proper technology.

On the sea floors and in the brine-filled waters themselves are vast storehouses of metals and minerals that can be used to help resolve resource shortages. However, "harvesting" the metals and minerals will require new technologies that do not disturb the fragile sea floor.

These are but a few of the massive ocean projects that can be explored. Perhaps more exciting are designs for Cities in the Sea.

Cities in the Sea

Colonization of the oceans is one of the last frontiers remaining on Earth. Prodigious oceanic city communities are inevitable and will be among the greatest achievements of a new society.

To fully utilize this bountiful wellspring of resources, we must develop large marine structures to explore the relatively untapped riches of the world's oceans. They will provide improved mariculture, fresh water production, power, and mining, which will offset land-based mining shortages. The oceans can provide almost unlimited riches in pharmaceuticals, chemicals, fertilizers, minerals, oil, natural gas, sweet water, and tidal and wind power, to name a few. Ocean-based and space-borne sensors would constantly track tidal flow, marine life, water composition and temperature, atmospheric conditions, and myriad other vital signs.

The development of these ocean communities would greatly relieve land-based population pressures. The population of such cities could vary from

several hundred to many thousand and they could be located throughout the world. They would be controlled, managed, and operated primarily by automated systems, and would be part of the international communications network. The oceans are, after all, essential to our survival and are a critical part of Earth's carrying capacity.

Use

Some of these cities could serve as universities and research centers where students from all nations could study marine sciences and management. They could also serve as monitoring stations of ocean currents, weather patterns, marine ecology, pollution, and geologic phenomena. For additional marine exploration, robotic submersibles would be designed and made available to everyone.

Other sea platforms could be used as rocket-launching systems. Space vehicles launched at the equator would save energy because the equator is the fastest moving portion of the earth. Locating launch sites there would take full advantage of Earth's rotation for additional thrust, requiring fewer thruster burns to reach geocentric orbit (the orbit where a satellite rotates with Earth and remains in a stationary position relative to it). For polar orbits, the launch platforms would be located off the west coast of the U. S. with computerized control and command systems located on ships or on the platforms themselves.

Not all areas of the oceans need to be used for technological development. Vast areas can be set aside for reclamation, enhancement, and preservation, making them a priority for global conservation.

For instance, the Caribbean and the Emerald Shoals of the vast banks of Eleuthera feature some of the clearest waters in the Bahamas and one of the most beautiful coral atolls in the Western Hemisphere. The waters surrounding these islands vary in hue from the magnificent deep blue of the Gulf Stream to shimmering shades of green. Similar areas exist in the South Pacific and many other locations throughout the world, where thousands of miles of shoreline remain unmarred by human habitation. In a new spirit of world cooperation, many of these areas can be set aside as international marine parks for the education and enjoyment of all. In these areas the only human intervention is to preserve and protect aquatic sanctuaries.

Life Styles on the Sea Cities

Future cities of the sea offer new and fascinating lifestyles for millions of inhabitants and are a favorite destination for all. Some would serve as underwater international parks where visitors observe the great protected reefs of the world. Through huge undersea windows they would be able to view the wonders of this environment in leisure and comfort; from a computerized chair, they could communicate with dolphins and other forms of marine life. Diving expeditions can be made through airlocks, and people are able to participate in research, sailing, scuba diving, and all the amenities that sea cities offer along with many other surface and sub-surface water activities -- without disturbing the balance of the marine environment.

Construction

Massive ocean structures would exist both above and beneath the sea. These structures would represent a spectacular engineering achievement with aircraft, sea craft, and submersible access. One of the most efficient designs would be a circular configuration, multi-storied, and fabricated of steel, using glass of superior strength and pre-stressed concrete reinforced with carbon fibers.

Some would be floating while others would be built on pilings with flotation barriers to prevent wind and heavy seas from damaging the structures. In deeper waters the floating platforms could be anchored to the seabed. Other ocean platforms could float freely, being self-propelled and extremely stable, ballasted by columns about 20 feet in diameter that penetrate 150 feet below the surface. To keep the platforms steady in any type of weather, the lower portions of these floating, cylindrical columns would contain a series of disks that extend out about six feet, spaced approximately ten feet apart. A belt surrounding the entire project would act as a breakwater.

Some of these cities can be constructed in technically developed countries and towed to their destination in sections, or as complete operating systems, similar to the manner in which oil platforms are transported to their destinations now. Other configurations would be variable composite structures, assembled on site and modified to serve many different functions, with the ability to be disassembled and relocated if required.

Other above-surface structures anchored to the seabed would serve as efficient bases for mining operations. These dome-shaped structures could

be almost totally automated, their flotation levels adjusted by flooding or emptying buoyancy chambers. They would be constructed in dry dock, towed to their destination, and then submerged and anchored into place. A floating dock system, which rises and falls with the tides and accommodates both surface and submersible craft, could be part of this design.

All marine development must be in full accord with the total carrying capacity and sustainability of the ocean environment. In the future, before any of these projects are built, the designers will account for the possible negative impact on the entire hydrosphere – the rivers, estuaries, lakes and oceans.

Energy

On these and other floating cities or platforms, powerful wind turbines can capture the ocean breezes. Solar and wind power generators are located on most of the upper decks. Cold water from the depths of the ocean can also be pumped up for uses such as the conversion of temperature differences into electrical energy. This process would provide a continuous supply of electricity far in excess of the cities' needs.

Mariculture

Mariculture, the planned cultivation of marine crops, and fish farming communities, can be designed to support more than one type of marine life. A mutually supporting symbiotic relationship can be sustained while emulating natural conditions as closely as possible. A wide variety of aquatic plants may be cultivated in multiple layers and suspended by cables in underwater fields adjacent to the cities. In some instances, the tops of plants could be harvested automatically, leaving the roots and lower third of the plant to grow new crops without replanting.

These floating ocean platforms would be equipped with solar-operated desalinization plants, which would extract fresh water for hydroponics farming and other uses. Upwelling can also be harnessed to extract deep-sea nutrients to supply aquaculture farming. Of course, any attempt at aquaculture or mariculture would be subject to international monitoring of ocean farms.

This provides fish farming complexes and introduces the most advanced principles of poly-culture, which maintains the reproduction and natural balance of species. Every precaution would be taken to avoid disrupting

or spoiling the spawning grounds that have sustained the human race for centuries.

Transportation

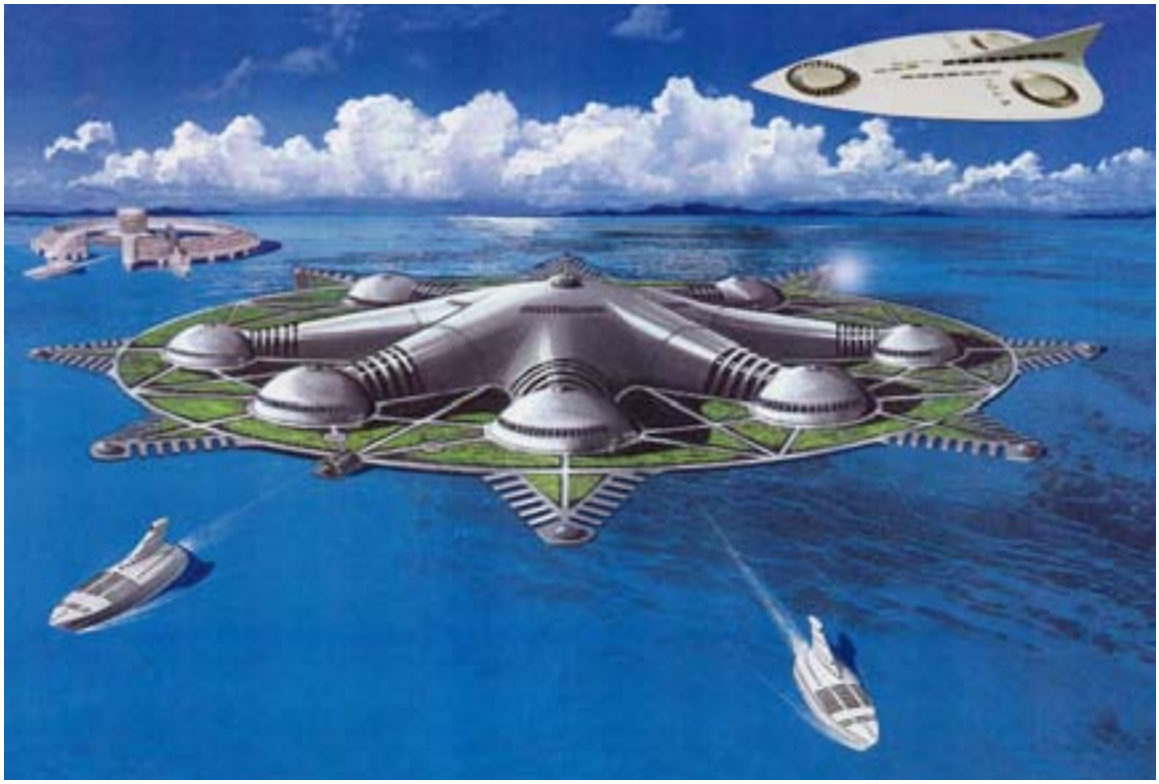
Immense floating structures can be equipped with loading and docking facilities for vessels. Huge ships that serve as processing plants could transport passengers and freight to these cities in the sea as well.

The upper deck of the sea cities would have a landing area for helicopters or VTOL aircraft. Computerized lift-units would facilitate vertical, horizontal, and radial travel within these structures.

Joint Venture

Where a project of this magnitude is concerned, like other resources, it is imperative that benefits be shared equally by the entire global community. The mineral wealth of the oceans and the other resources of our world must be shared by all nations as the common heritage of humankind.

Artificial Islands in the Sea



This artificial island in the sea is designed to serve the oceanographic sciences. Multiple docking and landing facilities for VTOL aircraft surround the entire island. Water-based recreation is part of life in these water-borne communities. People can participate in research, sailing, scuba diving, and many other surface and sub-surface activities without disturbing the balance of the marine environment.



Cities in the Sea

From the tops of these structures, a cylindrical concrete conduit extends 150 feet above the ocean surface. At surface level, it is encircled by a floating dock system, which rises and falls with the tides and accommodates both surface and submersible craft.

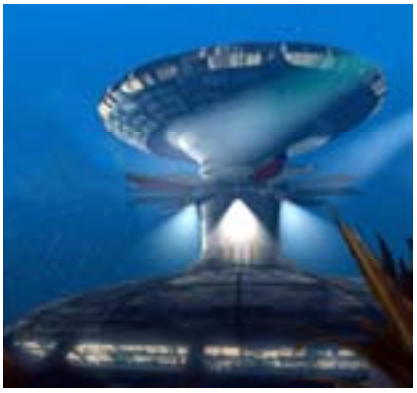


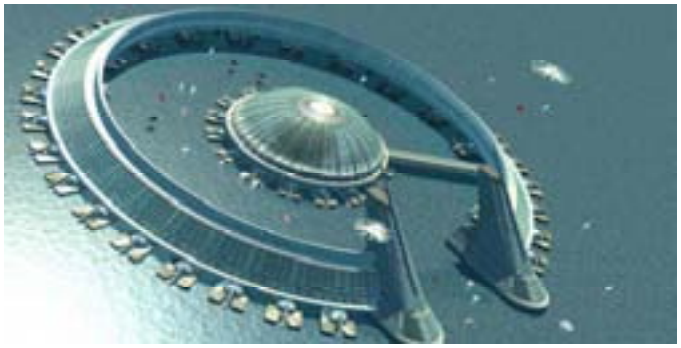
Cities in the Sea

Thousands of self-sufficient cities in the sea, varying in design according to their location and function, alleviate land-based population pressures. Some serve as oceanographic universities to survey and maintain a dynamic balance in the oceanographic environment.











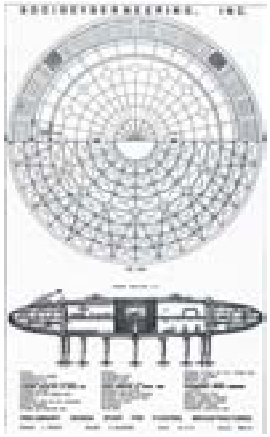




Ocean Mining Mega-Structures

These "cities in the sea" provide improved mariculture, fresh water production, power, and deep-ocean mining, which helps alleviate shortages of land-based minerals. Such structures provide us with almost unlimited resources in pharmaceuticals, chemicals, fertilizers, minerals, metals, oil, natural gas, drinking water, and ocean farming, as well as tidal and wind power and more.



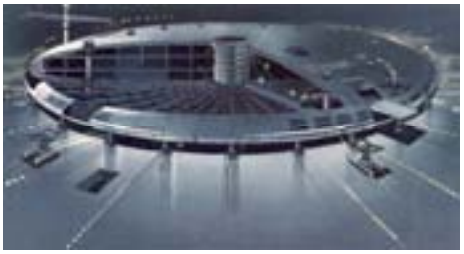


Construction of a Floating Mega-Structure

These illustrations depict a "floating mega-structure" that is being assembled in a dry dock entirely by robotized automated systems free of any human intervention. After construction, these structures are towed in sections or as completed units to locations where they are anchored to the seabed. In some instances, these cities can even travel and relocate.

City in the Sea With Modular Freighter





City in the Sea Modular Freighter



This modular freighter, leaving a city in the sea, consists of detachable sections that can be rapidly loaded or unloaded. The number of sections varies, depending on the amount of freight to be delivered. When all the modules are connected they are propelled as a single unit. When the freight arrives at its destination, the selected modules can be disconnected and towed to docks.

Apartments in the Sea Cities



Through huge undersea windows, occupants are able to view the wonders of this environment in leisure and comfort. From a computerized chair, they communicate with dolphins and other forms of marine life.

Floating Sea Domes



These unsinkable floating sea domes provide for those who prefer unique offshore or island living. In the event of severely inclement weather, they can easily be towed ashore, mounted, and locked to elevated support structures. They are equipped with retractable enclosures that cover the outer deck.

Solar Panels at Sea

On these and other floating cities or platforms, powerful wind turbines capture the ocean breezes. Solar and wind power generators are located on most of the upper decks and in the ocean as well.



Mariculture



Mariculture (the planned cultivation of marine crops) and fish farming communities are designed to support more than one type of marine life. Many of these communities maintain a balance of species in mutually supporting symbiotic relationships, while emulating natural conditions as closely as possible.



Chapter Ten

Decision Making and Laws

Decision Making

How are decisions arrived at in a cybernated resource-based society?

To answer this question, we use the scientific method and have computers get direct feedback from the environment. Computers would have electrical sensors extended into all areas of the social complex around the globe. For example, electrical sensors would extend into agricultural regions where computerized systems manage and control the agricultural requirements by monitoring the water table, insects, pests, plant diseases, soil nutrients, and so forth. Decisions are then arrived at using direct feedback from the environment. The result would be a more humane and meaningful civilization that is not based on the opinions or desires of a particular sect or individual.

One can think of this as a global autonomic nervous system. A human body reacts automatically to infection. If one has an infection in the toe there is no committee meeting of cells to inform the brain of the infection. The nervous system directs antibodies to the infected site automatically. This automatic response of the nervous system parallels the working of a resource-based economy.

The question we have to ask is "What end do we wish the culture to serve?" As the need for money is surpassed and the new mission is the well-being of all people and protection of the environment, the answers become readily available. The result is clean air and water, arable land, nutritious food, efficient transportation, a relevant education continuously updated, good health care, constructive social interaction, and cities functioning to serve these ends and more. This represents a more humane and meaningful approach for shaping civilization that is not based simply on opinion.

As we transition to a fully cybernated process of governing human affairs, newer technologies can be installed that remove human error. These machines would provide information rather than opinion, thus considerably reducing the influence of bias and irrational or purely

emotional elements in how affairs are managed. This way people play less and less of a role in decision-making and the society is working towards *AI* and machine decision-making to manage all resources, serving the common good.

Laws

Laws, at best, are attempts to control a population, and work only sporadically with great expense and hardship. Other common behavior-control methods are patriotism, religion, propaganda, and nationalism. All manmade laws are developed to preserve the established order. But laws never get to the root of problems, and they are continuously violated -- even by those who make them. When laws do not correspond to the nature of the physical environment, they will be violated. With so much economic deprivation and insecurity, even in the most affluent nations, no matter how many laws are enacted the same problems persist. In actuality, it is how the society is structured that is at fault.

Let's examine these concepts further.

The need for laws is a result of a *society being scarcity-oriented*. If a resource is abundant no one monitors it. When the necessities of life are abundant, monitoring is unnecessary.

In a resource-based economy social responsibility is not instilled by force, intimidation, or promises of heaven or threats of hell. Protection of the natural environment is not a matter of fines or penalties. Safeguards against abuse are designed into the environment. A simple example of this may be seen in the design of the cities where people have free access to resources without a price. This eliminates theft. Such measures are not a matter of passing and enforcing laws to prevent and punish abuse. Rather, they are a means for designing the flaws out of the social design, thus eliminating the need for many laws.

To eliminate traffic accidents a culture relying on the scientific method would not pass a law limiting speed to 55 mph. They would redesign the transportation systems so that accidents do not occur by using automated trains, monorails, transveyors (horizontal, vertical and radial transveyors), and individual units that have many sensors to diminish the possibilities of accidents.

A society with human concern "designs out" the need for laws and proclamations by making all things available to all people, regardless of race, color, or religious belief. When governments make laws, people are

led to believe that these laws are made to protect people's lives. In truth, laws are byproducts of insufficiency.

If we try to control human behavior by enacting laws or signing treaties without changing the physical conditions responsible for aberrant behavior, we are putting a Band-Aid on the issue. Instead of depending on a failed system of punishment or incarceration after the damage has been done, we should shift our attention to the inadequacies of society. These are things like poverty, malnutrition, homelessness, poor role models, a failed education, children lacking direction or interests, violence in the media, stresses in family life, and no positive vision for society to work toward.

A resource-based world economy brings about vast changes in human and interpersonal relations without the necessity of laws. It does this by introducing a set of values relevant to the needs of all people. It views all of the world's resources and technical information as the common heritage of all people. This is the unifying imperative. If accepted universally, the world will witness an end to the need for armaments, war, drugs, greed, and the other problems brought about by the endless pursuit of money and power.

Society has to understand that all of nature is subservient to natural law. Natural law cannot be violated without serious consequences to the individual or to society. Natural law dominates all living systems. For instance, without water, sun, or nutrients, plants and animals can not survive. Natural law is inviolable. A person who does not receive proper nutrition will not enjoy physical well-being and will soon become ill and die.

Similar Laws Apply to Human Behavior

Human behavior in all areas is just as subject to natural laws and the actions of external forces: it is generated by many interacting variables in one's own environment. This applies to behavior that is socially offensive. It is often influenced either by one's experiential background, nutritional factors in early life, or a number of other interrelated environmental factors.

When we see a dog leading a blind person across the street we tend to think it is a good dog. But when we see a dog bark at a cyclist we call it a bad dog. The dog is neither good nor bad. A dog can be trained to be ferocious or to help the blind. Both animals could be of the same breed, even from the same litter. Their different behavior is due to the differences in upbringing.

To put it another way, imagine an ancient Roman family watching Christians being fed to lions. Someone of today might be horrified and believe the people watching had trouble sleeping that evening. But they most likely had no trouble sleeping at all. Such bloodshed was the cultural sport of the times. Lions and Christians were looked upon with equal disdain.

Or imagine a modern day fighter pilot trained in warfare and taught a similar disregard for other cultures and beliefs losing sleep over shooting down twenty planes, and burning several inhabited villages. More likely, he will beam as he gets a medal and adorn his aircraft with symbols of his "kills." The pilot reflects his culture just as much as the Roman family does theirs. What we call our 'conscience' and 'morality' are not determined by an invisible "higher self." They are largely determined by geography, the times, and the individual's upbringing.

Whether they realize it or not, in the monetary system people are constantly manipulated through the media. People's most cherished beliefs are influenced by books, motion pictures, television, religions, role models, and the environment they live in. Even their notions of good and evil and their concepts of morality are part of their cultural heritage and experiences. This method of control does not require the use of physical force and is so successful we don't even recognize or feel the manipulation.

The dominant values of any social system rarely come from the people. Rather, they represent the views of the dominant control group such as the church, the military, the banks, the corporations, the power elite, or any combination thereof. These entities determine the public agenda, the courts, taxes, etc., all of which serve their own interests and perpetuate the illusion that society's values are determined from the ground up. In addition, governments suppress or explain away deviations that may threaten them.

With the scientific understanding that behavior is subject to the same natural laws that govern other processes, the educational system in the resource-based economy can evolve. It would teach process and analytical skills rather than rote memorization of facts. Dialogue would replace lecture. Understanding semantics is a skill that can greatly improve human communication and assist students in intelligently accessing relevant information. It is not that people will suddenly become better or more ethical, but that the conditions responsible for hostile and egocentric behavior would no longer exist.

If we want children to achieve a positive constructive relationship with one another and become contributing members of society, a way to accomplish this is by designing an environment that produces that desired behavior. For example, when the children are interested in learning how to assemble a small motor vehicle, the design can require four children to lift the car while two others attached the wheels. The rest of the car is assembled in a similar manner, needing the help and cooperation of everyone to complete the vehicle for use. This enlightened form of education helps students understand the advantages of cooperation.

Exercise would not be mandatory or monotonous, and it wouldn't involve adversarial competition. It would be incorporated into the learning experience. For example, a craft shop the children enjoy using is located on a hilltop in the middle of a lake. To get there the children have to row a boat, and then climb the hilltop. This not only provides exercise, but also a sense of achievement, which helps their mental health and increases incentive.

One of the greatest limiting factors in human systems is our inability to grasp the significance of underlying forces and the extent to which environment shapes our thinking, values, and/or behavior. When we speak of environment, we mean all of the interacting variables, which are the prime contributors to our mindset.

Chapter Eleven

Lifestyles

What Do People Do?

From early civilizations to the present day, most humans have had to work to earn a living. Most of our attitudes about work may be a carry-over from these much earlier times. In the past, it was necessary for people to fetch water and carry it to their dwelling places. They gathered wood to prepare fires for heating and cooking and fuel to burn in their lamps. It would have been very difficult for them to imagine a time when water would rush forth in their own home with the turn of a handle; to press a button for instant light would have seemed to be within the realm of magic. People of ancient times probably wondered what they would do with their time if they did not have to engage in the burdensome tasks that were so necessary to sustain their lives.

Alleviating Human Pressures

Human beings free of debt, insecurity, and fear become much more amiable. With no one out to sell anyone anything or to deprive another of possessions or money, the basis for unhealthy human aggression is outgrown. People no longer are burdened by the nagging concerns that consume so much attention such as mortgages, health care costs, education fees, fire insurance, economic recession or depression, the loss of jobs, and taxes. With the elimination of these burdens and the removal of the conditions that create feelings of envy, greed, and competition, people's lives would be far more meaningful.

The aim of this new social design is to encourage a new incentive system, one that is no longer directed toward the shallow and self-centered goals of wealth, property, and power. These new incentives encourage people toward self-fulfillment and creativity, the elimination of scarcity, the protection of the environment, and most of all concern for fellow human beings. People would have the means and the time for intellectual and spiritual growth, and the time to realize what it really means to be human in a caring society. Rather than evolving into an age of leisure, truly intelligent and committed people would find very little "free" time, even

without having to "work" for a living. There is always need for better more efficient ways of doing things. People will want to participate in the society, seeing how it directly benefits their way of life and others, as things are constantly being updated and changed. There are no Utopias. The very notion of "Utopia" is static. The survival of any social system ultimately depends upon its ability to allow for change to improve society as a whole. However, today, most people are not prepared for changes in society either emotionally or intellectually.

Unburdened by survival concerns, people would have time for individual interests such as continuing their education. Education, if available to everyone without a price tag, could become a never-ending process. The cities of the future would be living universities. Most people would attend schools and university classes or engage in other activities. There would be classes for those interested in theater, photography, painting, ballet, and all of the arts at the music, art, and theater centers. People would be able to go to centers to work on or get help with their new inventions.

New horizons open up for people who dared not even imagine these possibilities in the past due to lack of money or time. It is painful to even imagine a life of limitless possibilities for those who have limited purchasing power.

Today there are many boats at the docks but they are rarely used, let alone available to the majority of people. In this new society, there are more than enough boats provided for anyone's use to enjoy as they wish. There could be a surge in the number of people learning to fly. Imagine all the other numerous hobbies and sports -- the list of what would be available for people to pursue is endless.

Picture a world opened up for everyone's exploration, with many people traveling throughout the world, helping in lesser developed areas to bring them up to the highest living standards possible in the shortest time. A world where there is no begging for nickels and dimes for medical and scientific research -- there are major programs for research and development in the medical fields where people can also participate and learn. This process would be the same in all other areas of society.

Rather than specialize, most people would become generalists, learning more about many different disciplines and how they relate to one another. In this way, people would be better able to participate in the development of ideas in many areas of society. There would be so much expansion and experimental exploration in all areas that there would no longer be a concern about who decides which experiments are tried and

which are not. Every new design would be subject to exploration, testing, and evaluation, and those proven worthy would be built.

People would take advantage of opportunities to learn how to better communicate with others and resolve differences without violence. They would be introduced to tools for problem solving enabling them to participate in a wide variety of fields and explorations.

The notion of sitting idly by or retiring would become obsolete. People would have so many options that for the first time they could understand what it really means to be a member of a global society with not enough time in the day to do all that they would like.

Without the wasted time, resources, lives, and energy consumed by wars and preparations for war, our energy could be harnessed toward constructive endeavors. Society would be able to devote much more attention and resources to controlling unforeseen variables like tsunamis, earthquakes, hurricanes, and other natural disasters that threaten our lives. This does not imply perfection. There are always challenges and unresolved problems. But there would be a great improvement in life styles and the global society could achieve the highest standards possible for the times, both materially and spiritually.

How Resources are Distributed Equitably

In the world of tomorrow, distribution of goods and services can be accomplished without the use of money or tokens by way of large distribution centers. These centers would be similar to expositions where the advantages of new products are explained and demonstrated. Exhibition centers would display what is new and available and would be constantly updated.

Throughout the communities, there would be 3-D flat-screen imaging in each home. If you desire an item, an order would be placed and the item automatically delivered directly to your place of residence without a price tag, servitude, or debt of any kind. This would include whatever people need in the way of housing, clothing, education, health care, entertainment, etc.

Raw materials for products can be transported directly to manufacturing facilities by automated transportation "sequences" using boats, monorails, mag-lev trains, pipelines, and pneumatic tubes. An automated computerized inventory system would integrate the distribution centers and manufacturing facilities, coordinating production to meet demand.

In this way, a balanced-load economy can be maintained. Shortages, over-runs, and waste are eliminated in this system.

If a person visits Yellowstone National Park, they simply check out a camera or camcorder, use it (even choosing to attend a class to learn the features of the camera and photographic techniques), and then return it to another distribution center or drop-off, eliminating storage and maintenance. This is similar to a public library, but much more informative and inviting. There are so many interesting activities and learning areas at the arts and science centers that people can wander in and out of the sessions until they find something that relates to what they're looking for.

In this cybernated future, a couple may visit an architectural design center where they sit in front of a clear hemisphere approximately six feet in diameter. One person describes the type of house they would prefer and their areas of interest. The house appears as a 3-dimensional image in the center of the hemisphere. It rotates slowly to present an overview of the interior and exterior. Then the other person describes their major areas of interest and preferences and may suggest a larger balcony. The 3-dimensional image is adjusted accordingly. When they finish requesting changes, the computer presents various alternatives to consider. After deciding on all the alternatives they are able to enter a sensorium to experience a walk-through of their design and continue to make changes. When they arrive at the final design, the construction procedures are set into motion. The computer selects materials for efficiency and durability. None of the architecture is permanent and can be modified and updated at the request of the occupants. This is real individual choice.

In a monetary system most people live near their work with a house, car, and lifestyle they can afford (or, all too often, cannot afford), rather than one they prefer. They are essentially only as free as their purchasing power permits. Many wealthy people select a residence solely to impress others with their status. A resource-based economy changes the function of dwellings from that of status symbol or basic shelter to a reflection of individuality and personal interests.

Family Matters

While new technologies are quite amazing, you may well understand that the most profound effects are not in the technologies, but in our lifestyles

In most cases, our current system requires both husbands and wives to work. Monetary economics undermines family cohesion and childcare.

Parents lack adequate time for their children, and are constantly stressed by ever-rising medical bills, insurance payments, educational expenses, and cost of living expenses.

It is in this area that one of the most profound benefits of our new civilization shows. Shorter workdays would provide greater opportunities for family members to pursue areas of personal interest. Free access to goods and services makes home a more pleasant place, and removal of economic stress will reduce family turmoil. Society is designed so people are free to choose their own interests, develop formerly hidden potential, and pursue dreams without government intervention or financial constraint.

Conclusion

The conflicts today with our fellow human beings are over opposing values and limited access to the necessities of life. If we manage to arrive at a saner future civilization, the conflicts will be against problems common to all humans. In a vibrant and emergent culture, rather than having conflicts between nations, the challenges we will face will be overcoming scarcity, restructuring damaged environments, creating innovative technologies, increasing agricultural yield, improving communications, building communications between nations, sharing technologies, and living a meaningful life.

People would be free to pursue whatever constructive endeavors they chose without economic pressures, restraints, and taxation that are inherent in the monetary system. By constructive endeavors, we mean anything that enhances the lives of the individual and others. With these major alterations, people would eventually live longer more meaningful and healthier lives. The measure of success would be the fulfillment of one's individual pursuits rather than the acquisition of wealth, property, and power.

As we enhance the lives of others, protect our environment, and work toward abundance, all our lives can become richer and more secure. If these values were put into practice, it would enable all of us to achieve a much higher standard of living within a relatively short period of time; a standard of living that would be continuously improved. When education and resources are available to all without a price tag, there will be no limit to human potential.

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